[63/104] SEAT No.________

SC

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Sardar Patel University

F.Y.B.Sc. (CA & IT) (2nd Semester - NC) - CBCS Examination 2019

PS02CIIT03 || Advanced C Programming and Introduction to Data Structures

Date: 28-03-2019 , Thursday

Time: 10.00 am to 1.00 pm **Total Marks: 70**

Q.1 Answer the following questions. **[10]**

1 Which operator is used with a pointer to access the value of the variable whose address is contained in the pointer?

A. Address (&) B. Indirection (*) C. Assignment (=) D. Selection (->;)

2 Which of the following defines a pointer variable to an integer?

A. int &ptr; B. int **ptr; C. int *ptr; D. int &&ptr;

3 Which of the following is not a C memory allocation function?

A. malloc() B. calloc() C. realloc() D. alloc()

4 Which of the following allows a portion of memory to be shared by different types of data?

A. Array B. Union C. Structure D. File

5 f = fopen(filename, “r”);

Referring to the code above, what is the proper definition for the variable f?

A. FILE f; B. struct FILE f; C. FILE *f; D. int f;

6 When new data are to be inserted into a data structure, but there is no available space; this situation is usually called?

A. underflow B. housefull C. overflow D. saturated

7 The term PUSH and POP is related to the __________?

A. Array B. stacks C. queue D. All of these

8 Files are a ____________________type of Data Structure.

A. Linear B. Non-Primitive C. Primitive D. Non-Linear

9 A data structure that contains not only a data field but also contains pointer field is known as _________.

A. Queue B. Tree C .Stack D. Linked List

10 Which of the following is TRUE for a Queue data structure?

A. Linear B. Non- primitive C Both A and B D. None of the Above

Q.2 Explain following in brief. (Attempt any Ten) **[20]**

1 Differentiate: structure and union

2 List out benefits of pointers.

3 What is scale factor? Explain with example in brief.

4 List file modes available to manage the file in C.

5 Explain the fclose() function with example.

6 Explain typedef in brief with suitable example.

7 Give the examples of Primitive Data Structure.

8 State various Applications of Stack.

9 List out different applications of data Structure.

10 State various Applications of Linked List.

11 State various types of queue.

12 Define : Circular Queue and Priority Queue.

(P.T.O.)

Q.3	A.	Write a note on Dynamic memory allocation.	[05]
	B.	Explain pointer to structure array using appropriate example.	[05]
		OR	
Q.3	A.	Write note on: structure within structure	[05]
	B.	Explain pointer arithmetic with example.	[05]
Q.4	A.	Explain the getw() and fprintf() function with example.	[05]
	B.	What is union? Explain its definition, declaration and assigning values to members of union.	[05]
		OR	
Q.4	A.	Explain fscanf() and putw() function with example.	[05]
	B.	Describe the usage and limitation of getc() and putc() function.	[05]
Q.5	A.	Write a short note on primitive and non primitive data structure operations.	[05]
	B.	Write an algorithm for push and peep operation of a stack	[05]
		OR	
Q.5	A.	Write a short note on linear and non linear data structure.	[05]
	B.	Write an algorithm for pop and change operation of a stack	[05]
Q.6	A.	What is Linked List? Write an algorithm to insert an element into a Singly linked list that maintains ascending order of elements.	[10]
		OR	
Q.6	A.	What is queue? Write an algorithm to insert and delete an element into a simple queue.	[10]

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